

Technology News

Independent robotic warriors

Smart military robots controlled by humans are currently deployed in Pakistan to combat Taliban. Some like QinetiQ's MAARS are armed with weapons to shoot insurgents, appendages to disarm bombs and surveillance equipment to search buildings. No matter where the robots are deployed, however, there is always a human involved in directing where a robot should fly and what munitions the robot should use if it encounters resistance.

What if humans are taken out of the loop, and robots are left to decide who to kill or what to bomb on their own? Ronald Arkin, a Georgia Tech computer science professor, is develop-

ing an ethical governor—a package of software and hardware that tells robots when and what to fire. He argues not only can robots be programmed to behave more ethically on the battlefield, they may actually be able to make wiser decisions than human.

Arkin is translating the codified, written military law into terms that robots can understand and interpret themselves. One possible scenario for Arkin's ethical governor is an enemy sniper posted in a building next to an important cultural setting, like a mosque. A wheeled military robot emerges from cover and the sniper fires on it. The robot finds the sniper and has to choose between using a grenade launcher or its own sniper

rifle to bring down the fighter. Using geographical data on the surrounding buildings, the robot would decide to use the sniper rifle to minimise any potential damage to the surrounding buildings.

Robot goes social on Facebook

Can man and machine become friends? What if you receive a friend request from a robot at the social networking site Facebook? A robot is being given its own Facebook profile page in a bid to bridge the divide between humans and robots. Named and modeled after Arabic scholar Ibn Sina aka Avicenna, the robot interacts with humans. The interactions will be logged on its Facebook profile.

The coupling of robot and social network is the idea of Dr Nikolaus Mavridis and co-researchers.

Cool car technologies that face legal issues

Cars exist today that can steer themselves, brake to prevent collisions, run on electric power only, fly in the air and double as boats on the water. But, you can't have these cool technologies anytime soon as they are impractical for legal or financial reasons.

The new Toyota Prius, for example, uses its electric power steering to turn the steering slightly to keep the car in the lane if you start drifting out of it. Other makes, such as Volvo and Nissan, don't turn the steering, but suspend the power assist, except in the correct direction to keep the car in the lane, or they may selectively apply the brakes (as they can do with electronic stability control) to guide the car into its lane.

The fear is that the people could misuse automatic steering, and run over pedestrians while texting on mobile, for instance. And rather than



QinetiQ's MAARS robot